# Mathematical Induction, and Recursive Definitions.

## Mathematical Induction :-

Method of Mathematical Induction is based on a principle called the Induction Principal.

## Induction principle :-

This states as follows.

Let $S(n)$ denote an open statement that involves a positive integer $n$.

Suppose that the following conditions hold:

(1) $S(1)$ is true

(2) If whenever $S(k)$ is true for $k \in Z^+$, then $S(k+1)$ is true.

Then $S(n)$ is true for all $n \in Z^+$

## Method of Mathematical Induction :-

Suppose we wish to prove the statement $S(n)$ is true for all integers $n \geq 1$.

The method of proving such a statement based on induction principle is called the method of mathematical induction.

Scanned by Scanner Go

$S(K+1): 1+2+3+---+K+(K+1) = \frac{1}{2}(K+1)(K+2)$

Add $(K+1)$ to both sides

$(1+2+3+---+K)+(K+1) = \frac{1}{2}K(K+1)+(K+1)$

$= (K+1)\left(\frac{1}{2}K+1\right)$

$= (K+1)\left(\frac{K+2}{2}\right)$

$(1+2+3+---+K+K+1) = \frac{1}{2}(K+1)(K+2)$

we proved $S(n)$ is true for $n = K \geq 1$

(2) P.T, for each $n \in Z^+$

$1^2+2^2+3^2+---+n^2 = \frac{1}{6}n(n+1)(2n+1)$

Basis step: prove $S(1)$ is true

$1^2 = \frac{1}{6}(1)(1+1)(2(1)+1)$

$1 = \frac{1}{6}(2)(3)$

$1 = 1$

Induction step: prove $S(n)$ is true for $n = K$ where $K \geq 1$.

Assume $S(K)$ is true

$S(K): 1^2+2^2+---+K^2 = \frac{1}{6}(K)(K+1)(2K+1)$

add $(K+1)^2$ to both sides

Scanned by Scanner Go

$$= \frac{1}{6} k(k+1)(2k+1) + (k+1)^2$$

$$1^2+2^2+3^2+\cdots+k^2+(k+1)^2 = \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$= (k+1)\left\{\frac{k(2k+1)}{6} + (k+1)\right\}$$

$$= (k+1)\left(\frac{2k^2+k+6k+6}{6}\right)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3)$$

(3) Prove by mathematical induction, that

$$1^2+3^2+5^2+\cdots+(2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

for all integers $n \geq 1$.

(Solve)

(4) P.T

$$1\cdot2+2\cdot3+3\cdot4+\cdots+n(n+1) = \frac{1}{3}n(n+1)(n+2)$$

(Solve)

Scanned by Scanner Go

5) Prove by mathmatical induction that, for every +ve integer n, 5 divides $n^5 - n$

→ Let $S(n)$ : 5 divides $n^5 - n$

Basis step : prove $S(n)$ is true for $n=1$

i.e prove $S(1)$ is true

$S(1)$ : 5 divides $1^5 - 1$

$S(1)$ : 5 divides 0 which is true

so $S(1)$ is true

Induction step : Assume $S(n)$ is true for $n=k$ where $k \geq 1$.

$S(k)$ : 5 divides $k^5 - k$. → This means $k^5 - k$ is multiple of 5

$S(k)$ : $k^5 - k = 5m$ is true

$S(k+1)$ : $(k+1)^5 - (k+1)$

: $(k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1) - (k+1)$

: $k^5 + 5k^4 + 10k^3 + 10k^2 + 5k - k$

: $k^5 - k + 5(k^4 + 2k^3 + 2k^2 + k)$

: $5m + 5(k^4 + 2k^3 + 2k^2 + k)$

$S(k+1)$ : $5(m + k^4 + 2k^3 + 2k^2 + k)$

so proved $S(k+1)$ is also multiple of 5

so $S(n)$ is true for $n \geq 1$

Scanned by Scanner Go

(6) By mathematical Induction, prove that, for every +ve integer n, the number $A_n = 5^n + 2 \cdot 3^{n+1} + 1$ is a multiple of 8

Let $S(n)$ : $A_n = 5^n + 2 \cdot 3^{n+1} + 1$ is multiple of 8

Basis step: prove $S(1)$ is true for $n=1$

$$S(1): A_1 = 5^1 + 2 \cdot 3^{(1+1)} + 1$$
$$= 5 + 2 \cdot 3^2 + 1$$
$$= 5 + 2 + 1$$
$$\boxed{A_1 = 8}$$

proved $A_1$ is multiple of 8

Induction step: Assume $S(K)$ is true for $K \geq 1$

i.e $A_K$ is multiple of 8

we need to prove $A_{K+1}$ is multiple of 8.

So Take $A_{K+1} - A_K = \left(5^{K+1} + 2 \cdot 3^{K+1-1} + 1\right) - \left(5^K + 2 \cdot 3^{K-1} + 1\right)$

$$= 5^{K+1} + 2 \cdot 3^{K+1-1} + 1 - 5^K - 2 \cdot 3^{K-1} - 1$$
$$= 5^K(5-1) + 2 \cdot 3^{K-1}(3-1)$$
$$= 5^K(4) + 2 \cdot 3^{K-1}(2)$$
$$= 5^K \cdot 4 + 4 \cdot 3^{K-1}$$
$$A_{K+1} - A_K = 4\left(5^K + 3^{K-1}\right)$$

Take any K value $\in Z^+$ for which $5^K + 3^{K-1}$ is even. So $4(5^K + 3^{K-1})$ is multiple of 8.

So $A_{K+1} = \underbrace{4(5^K + 3^{K-1})}_{\text{multiple of 8}} + \underbrace{A_K}_{\text{multiple of 8}}$ is also multiple of 8

So An is true for some $n \in Z^+$.

Scanned by Scanner Go

This Method consists of 2 steps.

(1) Basis step : verify that the statement $S(1)$ is true i.e verify that $S(1)$ is true for $n=1$.

(2) Induction step : Assuming that $S(k)$ is true, where $k$ is an integer $\geq 1$, show that $S(k+1)$ is true.

Problems :-

(1) Prove by mathematical Induction that, for all +ve integers $n \geq 1$,

$$1+2+3+\cdots+n = \frac{1}{2}n(n+1)$$

we have to prove the statement

$$S(n) : 1+2+3+\cdots+n = \frac{1}{2}n(n+1)$$

for all integers $n \geq 1$

Basis step :

$$S(1) : \frac{1}{2} \cdot 1 \cdot (1+1)$$

$$S(1) : 1 = 1$$

$S(n)$ is true for $n=1$.

Induction step :

$$S(k) : 1+2+3+\cdots+k = \frac{1}{2}k(k+1)$$

Scanned by Scanner Go

(7) By mathematical induction, prove that $(n!) \geq 2^{n-1}$ for all integers $n \geq 1$.

$\longrightarrow$ $S(n) : (n!) \geq 2^{n-1}$

Basis step: prove $S(n)$ is true for $n = 1$

$S(1) : (1!) \geq 2^{1-1}$

$S(1) : 1 \geq 2^{0}$

$S(1) : 1 \geq 1$

Induction step: Assume $S(n)$ is true for $n = k$ i.e $S(k)$ is true i.e

$S(k) : k! \geq 2^{k-1}$ or $2^{k-1} \leq k!$ is true

$S(k+1) : 2^{(k+1)-1} = 2^{k} = 2 \cdot 2^{k-1}$

$\leq 2 \cdot k!$ → because $2 \geq k+1$ for $k \geq 1$

$\leq (k+1) \cdot k!$

$S(k+1) : 2^{k} \leq (k+1)!$

So $S(n)$ is true for $n = k+1$

Hence, by Mathematical Induction, $S(n)$ is true for all integers $n \geq 1$

Scanned by Scanner Go

(8) P.T $4n < (n^2 - 7)$ for all +ve integers $n \geq 6$

→ $S(n) : 4n < (n^2 - 7)$

Basis Step: prove $S(n)$ is true for $n = 6$

$S(6) : 4(6) < (6^2 - 7)$

$S(6) : 24 < 29$ which is true

Induction step: Assume $S(n)$ is true for $n = k$ i.e $S(k)$ is true.

$S(k) : 4k < (k^2 - 7)$

Prove $S(k+1)$ is true

$S(k+1) : 4(k+1) = 4k + 4$

$< (k^2 - 7) + 4$

$< (k^2 - 7) + (2k+1)$

↑ [BC2 $[2(6)+1] > 4$ when $k = 6$]

Note: Required Ans is $(k+1)^2 - 7$
i.e $k^2 + 2k + 1 - 7$ is required to get the Answer.
$k^2 - 7$ is already in the line above so replace 4 by $2k+1$ bcz $(2k+1) > 4$.

$< k^2 + 2k + 1 - 7$

$S(k+1) : 4(k+1) < (k+1)^2 - 7$

$S(k+1)$ is true

So $S(n)$ is true for all +ve integers $n \geq 6$.

Scanned by Scanner Go

(9) S.T $2^n > n^2$ for all +ve integers $n > 4$.

→ $S(n) : 2^n > n^2$

Basis step: prove $S(n)$ is true for $n = 5$

$S(5) : 2^5 > 5^2$

$S(5) : 32 > 25$ which is true

Induction step: Assume $S(k)$ is true for $n = k$. where $k \geq 5$

$S(k) : 2^k > k^2$

prove $S(k+1)$ is true

$S(k+1) : 2^{k+1} = 2^k \cdot 2$

$> k^2 \cdot 2$

$> k^2 + k^2$

~~$> k^2 + (k \cdot k)$~~

$> k^2 + 2k + 1$ [Bcz $(2k+1) < k^2$ for $k = 5$]

$S(k+1) : 2^{k+1} > (k+1)^2$

$S(k+1)$ is true

so $S(n)$ is true for $n > 4$ by Mathematical Induction

Scanned by Scanner Go

10 Let $a_0 = 1$, $a_1 = 2$, $a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. Prove that $a_n \leq 3^n$ for all +ve integers $n$.

$$S(n) : a_n \leq 3^n$$

Basis Step: prove $S(n)$ is true for $n = 0, 1, 2$

$S(0) : a_0 \leq 3^0$ | $S(1) : a_1 \leq 3^1$ | $S(2) : a_2 \leq 3^2$

$1 \leq 1$ | $2 \leq 3$ | $3 \leq 9$

$S(n)$ is true for $n = 0, 1, 2$.

Induction step: Assume $S(n)$ is true for $n = k$.

$$S(k) : a_k \leq 3^k$$

Prove $S(k+1)$ is true

$$S(k+1) : a_{k+1} = a_{k+1-1} + a_{k+1-2} + a_{k+1-3}$$

$$= a_k + a_{k-1} + a_{k-2}$$

$$\leq 3^k + 3^{k-1} + 3^{k-2}$$

$$\leq 3^k + \underbrace{3^k + 3^k}_{\text{BC2 } 3^k > 3^{k-1}}$$

$$\leq 3 \times 3^k$$

$$\boxed{S(k+1) : a_{k+1} \leq 3^{k+1}}$$

$S(k+1)$ is true

so $S(n)$ is true for all +ve integers $n$.

Scanned by Scanner Go

(11) Prove that every +ve integer $n \geq 24$ can be written as a sum of 5's and 7's.

$S(n)$ : n can be written as a sum of 5's and/or 7's.

Basis step: prove $S(n)$ is true for $n = 24$

$$24 = (7+7) + (5+5)$$

24 represented in terms of 7's and/or 5's.

Induction step: Assume $S(n)$ is true for $n = k$

$S(k)$ : k can be written as a sum of 5's and/or 7's.

Prove $(k+1)$ can be written as a sum of 5's and/or 7's.

$$(k+1) = \{\underbrace{(7+7+--7)}_{r} + \underbrace{(5+5+5+--)}_{s}\} + 1$$

where $r \rightarrow$ number of 7's
$s \rightarrow$ number of 5's
where $r \geq 2$ & $s \geq 2$

$$= \underbrace{(7+7+--)}_{r-2} + (7+7) + \underbrace{(5+5+5+--)}_{s} + 1$$

$$= (7+7+--) + (5+5+--) + 7+7+1$$

$$= (7+7+--) + (5+5+--) + 15$$

$$\therefore (k+1) = (7+7+--) + (5+5+--) + 5+5+5$$

So $(k+1)$ is represented in terms of 7's and/or 5's

$S(n)$ is true for $n \geq 24$

Scanned by Scanner Go

(12) Prove by mathematical induction that, for any +ve integer n, the number $11^{n+2} + 12^{2n+1}$ is divisible by 133

→ $S(n) : 11^{n+2} + 12^{2n+1}$ is divisible by 133

Basis step: prove $S(n)$ is true for $n=1$

$S(1) : 11^{1+2} + 12^{2(1)+1}$ is divisible by 133

$: 11^3 + 12^3$

$S(1) : 3059$ is divisible by 133

133)3059(23
3059
00

Induction step: Assume $S(n)$ is true for $n=K$

$S(K) : 11^{K+2} + 12^{2K+1}$ is divisible by 133

prove $S(K+1)$ is true.

$S(K+1) = 11^{(K+1)+2} + 12^{2(K+1)+1}$

$= (11^{K+2} \times 11) + (12^{2K+2+1})$

$= (11^{K+2} \times 11) + (12^{2K+1} \times 12^2)$

$= (11^{K+2} \times 11) + (12^{2K+1} \times 144)$

$= (11^{K+2} \times 11) + (12^{2K+1} \times (11+133))$

$= (11^{K+2} \times 11) + (12^{2K+1} \times 11) + (12^{2K+1} \times 133)$

$= 11(11^{K+2} + 12^{2K+1}) + (12^{2K+1} \times 133)$

$= (11 \times \underbrace{AK}_{\text{multiple of } 133}) + (12^{2K+1} \times 133)$

So $S(K+1)$ is multiple of 133

$S(n)$ is true for $n \geq 1$

Scanned by Scanner Go

## Q.P Questions :- (Chapter 3)

(1) Prove by using mathematical induction

$$\sum_{i=1}^{n} i \cdot 2 = 2 + (n-1) \cdot 2^{n+1}$$

(2) P.T $4n < (n^2 - 7)$ for all integers $n \geq 6$

(3) Prove by mathematical induction that:

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3} n (2n-1)(2n+1)$$

(4) Prove that for any +ve integer n,

$$\sum_{i=1}^{n} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{n+2}}{2^n}$$

(5) Use Mathematical Induction to prove that 5 divides $n^5 - n$, where n is a non-negative integer.

(6) Prove the following, for all $n \geq 1$ using the principle of Mathematical Induction:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

(7) Prove by Mathematical Induction, for every +ve integer 8 divides $5^n + 2 \cdot 3^{n-1} + 1$

(8) Prove the following result by mathematical induction

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6} n (n+1)(2n+1)$$

Scanned by Scanner Go

(9) Prove that every +ve integer $n \geq 24$ can be written as sum of 5's and/or 7's.

(10) Prove that every +ve integer greater than or equal to 14 may be written as sum of 3's and/or 8's.

(11) If $L_0, L_1, L_2 \ldots$ are Lucas numbers, then prove that $L_n = \left[\frac{1+\sqrt{5}}{2}\right]^n + \left[\frac{1-\sqrt{5}}{2}\right]^n$

(12) For the Fibonacci sequence $F_0, F_1, F_2$ prove that $F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$

(13) $1*3 + 2*4 + 3*5 + \ldots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$

(14) The Lucas numbers are defined recursively by $L_0 = 2$, $L_1 = 1$, and $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$. Evaluate $L_2$ to $L_{10}$

(15) Prove by mathematical Induction that for all integers $n \geq 1$.

$$1 + 2 + 3 + \ldots + n = \frac{1}{2}n(n+1)$$

(16) The fibonacci numbers are designed recursively by $F_0 = 0$, $F_1 = 1$, ($F_n = F_{n-1} + F_{n-2}$) for $n \geq 2$. Evaluate $F_2$ to $F_{10}$

(17) By Mathematical Induction prove that for any +ve integer n the number $11^{n+2} + 12^{2n+1}$ is divisible by 133

Scanned by Scanner Go

# Recursive Definitions

Sequence → It is an arrangement of elements in order.

ex:- ① $a_1, a_2, a_3 \text{ -- } a_n$ is a sequence in which $a_1$ is the first element, $a_2$ is the second element. -- $a_n$ is the $n^{th}$ element.

Above sequence is denoted by $a_n$ ($n^{th}$ term)

② $a_0, a_1, a_2 \text{ -- } a_{n-1}$ is also a sequence in which $a_0$ is the I$^{st}$ element, $a_1$ is the II element --- $a_{n-1}$ is the $n^{th}$ element.

Above sequence is denoted by $a_{n-1}$ ($n^{th}$ term)

Sequence is described using 2 Methods

① Explicit Method
② Recursive Method

In Explicit Method, the general term of the sequence is explicitly indicated.

In recursive method, first few terms of the sequence are explicitly indicated & the general term is specified through a formula from which new terms can be obtained with the help of known terms.

Scanned by Scanner Go

Example :- Sequence E formed by all +ve even integers.

(a) Explicit method of specifying the above sequence.

$E = \{a_n\}$ where $a_n = 2n$, $n \in Z^+$

$a_n = 2n$ : general term of the sequence specified explicitly.

(b) Recursive method of specifying the above sequence

$E = \{a_n\}$, where $a_1 = 2$ & $a_n = a_{n-1} + 2$ for $n \geq 2$

$a_1 = 2$ : I term is specified explicitly

$a_n = a_{n-1} + 2$ : Terms after the I term are specified by a formula which indicates how to obtain a new term from the preceding term.

Scanned by Scanner Go

Problems :-

① Obtain a recursive definition for the sequence $\{a_n\}$ in each of the following cases.

(a) $a_n = 5n$

→ $a_1 = 5$, $a_2 = 10$, $a_3 = 15$, $a_4 = 20$, ----

so $\boxed{a_1 = 5, \text{ and } a_n = a_{n-1} + 5 \text{ for } n \geq 2}$

(b) $a_n = 6^n$

→ $a_1 = 6^1$, $a_2 = 6^2$, $a_3 = a^3$, ----

so $\boxed{a_1 = 6 \text{ and } a_n = a_{n-1} \times 6 \text{ for } n \geq 2}$

or

$\boxed{a_1 = 6 \text{ and } a_{n+1} = a_n \times 6 \text{ for } n \geq 1}$

(c) $a_n = 3n + 7$

→ $a_1 = 10$, $a_2 = 13$, $a_3 = 16$, $a_4 = 19$, ----

so $\boxed{a_1 = 10 \text{ and } a_n = a_{n-1} + 3 \text{ for } n \geq 2}$

(d) $a_n = n(n+2)$

$a_1 = 3$, $a_2 = 8$, $a_3 = 15$, $a_4 = 24$, ----

so $\boxed{a_1 = 3 \text{ and } a_n = a_{n-1} + 2n + 3 \text{ for } n \geq 2}$

(e) $a_n = n^2$

(solve)

Scanned by Scanner Go

(f) $a_n = 2 - (-1)^n$

$\rightarrow a_{n+1} - a_n = \left[2 - (-1)^{n+1}\right] - \left[2 - (-1)^n\right]$

$= \{-[(-1)^n \times (-1)]\} + \{(-1)^n\}$

$= (-1)^n \{1 + 1\}$

$\boxed{a_{n+1} - a_n = 2(-1)^n}$

$a_1 = 3,\ a_2 = 1,\ a_3 = 3$ ____

so $\boxed{a_1 = 3 \text{ and } a_{n+1} = a_n + 2(-1)^n \text{ for } n \geq 1}$

(2) A sequence $\{a_n\}$ is defined recursively by $a_1 = 4,\ a_n = a_{n-1} + n$ for $n \geq 2$. Find $a_n$ in explicit form.

$\rightarrow$ given $a_n = a_{n-1} + n$

$= [a_{n-2} + (n-1)] + n$

$= [a_{n-3} + (n-2)] + (n-1) + n$

$= [a_{n-4} + (n-3)] + (n-2) + (n-1) + n$

add (+n) & (-1) HOW $\longrightarrow$ $= a_1 + 2 + 3 + 4 + \text{---} + n$

$= (a_1 - 1) + (1 + 2 + 3 + \text{---} + n)$

given $\rightarrow (a_1 = 4)$ $= (4 - 1) + (1 + 2 + 3 + \text{---} + n)$

Scanned by Scanner Go

w.k.t $1+2+3+4+\cdots+n=\frac{1}{2}n(n+1)$

so $\boxed{a_n = 3+\frac{1}{2}(n)(n+1)}$

↓ This is the explicit formula for $a_n$.

---

(3) Find an explicit definition of the sequence defined recursively by

$a_1=7,\ a_n=2a_{n-1}+1$ for $n\geq 2$

$\rightarrow a_n = 2a_{n-1}+1$

$= 2\{2a_{n-2}+1\}+1$

$= 2[\{2(2a_{n-3}+1)+1\}]+1$

$= 2^3 a_{n-3} + 2^2+2+1$

$\vdots$

$= 2^{n-1}a_{n-(n-1)} + 2^{n-2}+2^{n-3}+\cdots+2^2+2+1$

$= 2^{n-1}a_1 + (1+2+2^2+2^3+\cdots+2^{n-3}+2^{n-2})$

Given $a_1=7$ & a standard formula

$1+a+a^2+\cdots+a^{n-1} = \frac{a^n-1}{a-1}$ for $a>1$

$a_n = 7\times 2^{n-1} + \left(\frac{2^n-1}{2-1}\right)$

$= (7\times 2^{n-1}) + (2^{n-1}-1) = 2^{n-1}(7+1)-1$

$\boxed{a_n = (8\times 2^{n-1})-1}$ ← This is the explicit definition

Scanned by Scanner Go

(4) A sequence $\{C_n\}$ is defined recursively by $C_n = 3C_{n-1} - 2C_{n-2}$ for all $n \geq 3$, with $C_1 = 5$ and $C_2 = 3$ as the initial conditions. Show that $C_n = (-2)^n + 7$

we have $n = 3$

$$C_3 = 3C_2 - 2C_1$$
$$= 3(3) - 2(5)$$
$$= 9 - 10 = -1$$

$$\boxed{C_3 = (-2)^3 + 7}$$

i.e expression $C_n$ is true for $n = 3$.

Assume $C_n$ is true for $n = 3, 4, 5 \ldots K$ and prove $C_n$ is true for $n = K+1$

$$C_{K+1} = 3C_{(K+1)-1} - 2C_{(K+1)-2}$$
$$= 3C_K - 2C_{K-1}$$
$$= 3[(-2)^K + 7] - 2 \times (-2^{K-1} + 7)$$
$$= (-3 \cdot 2^K + 21) + (2^1 \cdot 2^{K-1} + 14)$$
$$= (-3 \times 2^K + 21) + (2^K - 14)$$
$$= (-3+1) \cdot 2^K + (21 - 14)$$
$$= -2 \cdot 2^K + 7$$

$$\boxed{C_{K+1} = -2^{K+1} + 7}$$

This shows that the expression $C_n = -2^n + 7$ is true for $n = K+1$

Hence, by mathematical induction, the expression is true for all integers $n \geq 3$.

Scanned by Scanner Go

(5) A sequence $\{a_n\}$ is defined recursively by

$a_0 = 1$, $a_1 = 1$, $a_2 = 1$ and $a_n = a_{n-1} + a_{n-3}$ for all $n \geq 3$

prove that $a_{n+2} \geq (\sqrt{2})^n$ for all integers $n \geq 0$

I^st prove $a_3$ and $a_4$ are the addition of $(a_2 + a_0)$ and $(a_3 + a_1)$ because $a_n = a_{n-1} + a_{n-3}$

given $a_0 = 1$, $a_1 = 1$, $a_2 = 1$

$a_3 = a_2 + a_0 = 2$

$a_4 = a_3 + a_1 = 3$ and so on

Next prove that given $a_{n+2} > (\sqrt{2})^n$ is true for $n = 0, 1, 2$.

when $n = 0$, $a_{0+2} > (\sqrt{2})^0$

$\therefore a_2 > 1 = \boxed{1 > 1}$

when $n = 1$ $a_{1+2} > (\sqrt{2})^1$

$a_3 > \sqrt{2} = \boxed{2 > 1.41428}$

when $n = 2$ $a_{2+2} > (\sqrt{2})^2 = \boxed{3 > 2}$

Thus proved.

Next assume given statement is true for $n = 0, 1, 2, \ldots K$ for $K \geq 2$. Then,

$a_{K+3} = a_{K+2} + a_K$

$\geq (\sqrt{2})^K + (\sqrt{2})^{K-2}$

Scanned by Scanner Go

$$\geq (\sqrt{2})^{k+2-2} + (\sqrt{2})^{k-2}$$

$$\geq (\sqrt{2})^{k-2} \left((\sqrt{2})^2 + 1\right)$$

$$\geq (\sqrt{2})^{k-2} (2+1)$$

$$\geq 3(\sqrt{2})^{k-2}$$

For $a_{k+3}$ i should get $(\sqrt{2})^{k-1}$ so in the above line replace 3 by still lower value $[(\sqrt{2})^3 = 2\sqrt{2} = 2.828]$.

$$a_{k+3} \geq (\sqrt{2})^3 (\sqrt{2})^{k-2}$$

$$a_{k+3} \geq (\sqrt{2})^{3+k-2}$$

$$\boxed{a_{k+3} \geq (\sqrt{2})^{k+1}}$$

This shows that the statement is true for $n = k+1$.

So $a_{n+2} \geq (\sqrt{2})^n$ is true for $n \geq 0$

---

(6) The Lucas numbers are defined recursively by

$L_0 = 2$, $L_1 = 1$, and $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$

Evaluate $L_2$ to $L_{10}$

$L_2 = L_1 + L_0 = 1 + 2 = 3$

$L_3 = L_2 + L_1 = 3 + 1 = 4$

Scanned by Scanner Go

$L_4 = L_3 + L_2 = 4 + 3 = 7$

$L_5 = L_4 + L_3 = 7 + 4 = 11$

$L_6 = L_5 + L_4 = 11 + 7 = 18$

$L_7 = L_6 + L_5 = 18 + 11 = 29$

$L_8 = L_7 + L_6 = 29 + 18 = 47$

$L_9 = L_8 + L_7 = 47 + 29 = 76$

$L_{10} = L_9 + L_8 = 76 + 47 = 123$

---

(7) The Fibonacci numbers are defined recursively by

$F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

Evaluate $F_9$ to $F_{10}$

solve this

---

(8) For the Fibonacci sequence $F_0, F_1, F_2 \ldots$ prove that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

First, prove that $F_n$ is true for $n = 0, 1$ i.e prove $F_0 = 0$ and $F_1 = 1$ because in Fibonacci series I$^{st}$ 2 terms are 0 & 1.

$$F_0 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^0 - \left(\frac{1-\sqrt{5}}{2}\right)^0\right] = \frac{1}{\sqrt{5}}(1-1) = \boxed{0}$$

$$F_1 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^1 - \left(\frac{1-\sqrt{5}}{2}\right)^1\right] = \frac{1}{\sqrt{5}}\left[\frac{1}{2} + \frac{\sqrt{5}}{2} - \frac{1}{2} + \frac{\sqrt{5}}{2}\right]$$

$$= \frac{1}{\sqrt{5}}\left(\frac{2\sqrt{5}}{2}\right) = \boxed{1}$$ Hence proved for $n = 0$ & $1$

Scanned by Scanner Go

we assume that the result is true for $n = 0, 1, 2, \ldots\ldots k$ where $k \geq 1$.

$$F_{k+1} = F_k + F_{k-1}$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right] + \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left\{\frac{1+\sqrt{5}}{2}+1\right\} + \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left\{\frac{1-\sqrt{5}}{2}+1\right\}\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left\{\left(\frac{1+\sqrt{5}+2}{2}\right)\right\} + \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left(\frac{1-\sqrt{5}+2}{2}\right)\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left(\frac{3+\sqrt{5}}{2}\right) + \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left(\frac{3-\sqrt{5}}{2}\right)\right]$$

Divide & multiply by 2

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left(\frac{6+2\sqrt{5}}{4}\right) + \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left(\frac{6-2\sqrt{5}}{4}\right)\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left(\frac{1+\sqrt{5}}{2}\right)^2 + \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left(\frac{1-\sqrt{5}}{2}\right)^2\right]$$

$$\boxed{F_{k+1} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} + \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right]}$$

This shows that the given statement is true for $n = k+1$.

Scanned by Scanner Go

(9) For integers $m$ & $k$, the Eulerian numbers $a_{m,k}$ are defined recursively as follows:

$$a_{0,0} = 1$$

$$a_{m,k} = 0 \text{ for } k \geq m > 0$$

$$a_{m,k} = 0 \text{ for } k < 0$$

$$a_{m,k} = (m-k)\, a_{m-1,k-1} + (k+1)\, a_{m-1,k} \text{ for } 0 \leq k \leq m-1 \quad \text{—(i)}$$

Determine the values of $a_{m,k}$ for $1 \leq m \leq 5$ and $0 \leq k \leq m-1$.

→ when $m=1$ & $k=0$

$$a_{1,0} = (1-0)\, a_{1-1,0-1} + (0+1)\, a_{1-1,0}$$

$$= a_{0,-1} + a_{0,0}$$

$$\boxed{a_{1,0} = 0 + 1 = 1}$$

→ when $m=2$, & $k=0,1$

$$a_{2,0} = (2-0)\, a_{2-1,0-1} + (0+1)\, a_{2-1,0}$$

$$= 2 \cdot a_{1,-1} + a_{1,0}$$

$$= 2 \cdot 0 + 1$$

$$\boxed{a_{2,0} = 1}$$

$$a_{2,1} = (2-1)\, a_{2-1,1-1} + (1+1)\, a_{2-1,1}$$

$$= a_{1,0} + 2 \cdot a_{1,1}$$

$$= 1 + 2 \cdot (0)$$

$$\boxed{a_{2,1} = 1}$$

Scanned by Scanner Go

→ when $m=3$ and $K=0,1,2$

$$a_{3,0} = (3-0)a_{3-1,0-1} + (0+1)a_{3-1,0}$$
$$= 3 \cdot a_{2,-1} + a_{2,0}$$
$$= 3 \cdot (0) + 1$$

$$\boxed{a_{3,0} = 1}$$

$$a_{3,1} = (3-1)a_{3-1,1-1} + (1+1)a_{3-1,1}$$
$$= 2 \cdot a_{2,0} + 2 \cdot a_{2,1}$$
$$= 2 \cdot (1) + 2 \cdot (1)$$

$$\boxed{a_{3,1} = 4}$$

$$a_{3,2} = (3-2)a_{3-1,2-1} + (2+1)a_{3-1,2}$$
$$= a_{2,1} + 3 \cdot a_{2,2}$$
$$= 1 + 3 \cdot (0)$$

$$\boxed{a_{3,2} = 1}$$

Now we find from expression (i)

$$a_{4,3} = (4-3)a_{3,2} + (3+1)a_{3,3}$$
$$= a_{3,2} + 4 \cdot a_{3,3} = 1 + 4(0) = 1$$

$$a_{4,2} = (4-2)a_{3,1} + (2+1)a_{3,2}$$
$$= 2 \cdot 4 + 3 \cdot 1 = 11$$

$$a_{4,1} = (4-1)a_{3,0} + (1+1)a_{3,1}$$
$$= 3 \cdot 1 + 2 \cdot 4 = 11$$

$$a_{4,0} = (4-0)a_{3,-1} + (0+1)a_{3,0}$$
$$= 4 \cdot 0 + 1 \cdot 1 = 1$$

Scanned by Scanner Go

$a_{5,4} = (5-4)a_{4,3} + (4+1)a_{4,4}$

$= a_{4,3} + 5 \cdot a_{4,4} = 1 + 5 \cdot 0 = 1$

$a_{5,3} = (5-3)a_{4,2} + (3+1)a_{4,3}$

$= 2 \cdot a_{4,2} + 4 \cdot a_{4,3} = 2 \cdot 11 + 4 \cdot 1 = 26$

$a_{5,2} = (5-2)a_{4,1} + (2+1)a_{4,2}$

$= 3 \cdot a_{4,1} + 3 \cdot a_{4,2} = 3 \cdot 11 + 3 \cdot 11 = 66$

$a_{5,1} = (5-1)a_{4,0} + (1+1)a_{4,1}$

$= 4 \cdot a_{4,0} + 2 \cdot a_{4,1} = 4 \cdot 1 + 2 \cdot 11 = 26$

$a_{5,0} = (5-0) \cdot a_{4,-1} + (0+1) \cdot a_{4,0}$

$= 5 \cdot 0 + a_{4,0} = 0 + 1 = 1$

All the required values of $a_{m,k}$ are then computed.

Scanned by Scanner Go